USDX: A Dollar Built on the U.S. Mortgage Market

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Abstract

Stablecoins now move tens of trillions of dollars in annualized onchain transaction volume, with a combined market capitalization of approximately $322 billion as of May 2026 [1]. Yet the assets backing them remain narrowly concentrated in short-duration U.S. Treasuries and cash. This reserve composition mirrors the posture of money market funds, not the composition of a durable monetary system. The U.S. dollar, in its institutional form, is supported by a much broader asset base — and the largest single component of that base is not Treasuries but real estate debt. U.S. residential mortgage debt outstanding stands at approximately $13.2 trillion — a sum equal to well over half the entire $22.6 trillion U.S. M2 money supply — with the substantial majority securitized in agency mortgage-backed securities (MBS) and traded in one of the deepest fixed-income markets in the world [2]. So central is this asset class to the dollar that the Federal Reserve itself holds roughly $2.2 trillion of agency MBS on its own balance sheet. It underwrites a substantial fraction of the money supply through bank credit and Federal Reserve operations.

USDX is a USD-pegged stablecoin backed by a portfolio of U.S. residential mortgages and agency MBS. It brings the most-trusted collateral class in American finance onchain, giving any wallet access to the cashflows that have underwritten the American dollar for a century. Holders can stake USDX for mUSDX, a yield-bearing token that accrues the return of the underlying mortgage portfolio net of protocol fees. Peg integrity is maintained by direct redemption against reserves, repo lines against agency MBS for additional liquidity, and a decentralized arbitrage surface across liquidity pools. A separate token, RATES, serves two functions: validators stake RATES to verify loan data and property attestations that gate the minting of new USDX collateral, and RATES stakers provide a first-loss safety module against collateral shortfalls, creating a structural buffer between protocol risk and stablecoin holders.

This paper specifies the collateral framework, mint and redeem mechanics, yield distribution, peg stability apparatus, loss waterfall, technical architecture, and token system of USDX. USDX sources mortgage collateral both from secondary markets and from a native onchain origination channel, allowing the reserve to grow with newly originated loans rather than only through secondary purchases. That $13.2 trillion of debt is secured against roughly $50 trillion of U.S. residential real estate, leaving on the order of $34 trillion in untapped home equity — economically real wealth that today is largely inaccessible without selling or refinancing — and USDX is designed to bring this collateral base onchain. USDX is designed as infrastructure: a dollar backed by the same asset class that underwrites the institutional dollar, composable as a digital asset, and redeemable, auditable, and productive by construction.

Keywords: stablecoin, mortgage-backed securities, tokenized real estate, peg stability, collateralized debt, safety module, onchain finance.

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1. Introduction

The U.S. dollar, when measured by the assets that support it, is a claim on an extraordinarily broad economic base. Of the approximately $22.6 trillion M2 money supply as of early 2026, roughly $18 trillion resides in the commercial banking system as deposits backed by loans [3]. The single largest loan category on U.S. bank balance sheets is not commercial lending, consumer credit, or corporate debt, but residential mortgages. The Federal Reserve itself holds approximately $2.2 trillion of agency MBS on its balance sheet as of early 2026, making it the largest single mortgage investor in the country [4]. In aggregate, residential mortgages and agency MBS underwrite a substantial share of the institutional dollar.

Stablecoins, in their current form, do not inherit this breadth. Circle's USDC and Tether's USDT — the two largest dollar-pegged tokens by circulation — are backed predominantly by short-duration Treasury bills, overnight repo, and cash held at commercial banks. This is a valid design choice for money-market-like exposure, but it produces a different animal from the fully diversified dollar. Stablecoin reserves today look like the asset side of a government money market fund; the institutional dollar looks like the combined balance sheet of the banking system and the Federal Reserve.

This whitepaper proposes a stablecoin whose collateral composition more closely resembles the latter. USDX is backed by agency MBS, whole mortgage loans, tokenized mortgages, and liquid reserves. Its yield derives from mortgage coupon payments and MBS interest distributions — cashflows that have been the foundation of American housing finance for over fifty years. Its redemption guarantee is rooted in a combination of dollar reserves and repo facilities against MBS holdings, the same liquidity mechanism that the largest money managers and central banks rely on.

The asset class is chosen deliberately. U.S. household real estate is valued at approximately $47.6 trillion as of Q3 2025, against $13.2 trillion in mortgage debt outstanding, yielding roughly $34.4 trillion of homeowner equity — a figure that exceeds the market capitalization of every major public equity index [5]. Most homeowners cannot readily access this equity without refinancing or selling. On the institutional side, agency MBS trade in the To-Be-Announced (TBA) market, which is the second most liquid fixed-income market in the world after U.S. Treasuries. Bringing this collateral onchain addresses inefficiency at both ends of the market: it gives stablecoin holders exposure to an asset class from which they are currently excluded, and it extends the distribution of mortgage cashflows from a small set of institutional allocators to anyone, anywhere, with an internet connection — the largest potential distribution any financial asset has ever had.

The contributions of this paper are:

1. A collateral framework that specifies how agency MBS, whole loans, tokenized mortgages, and liquid reserves combine to back a dollar-pegged stablecoin, including allocation ranges and custody requirements.
2. A dual mint-and-redeem mechanism that accommodates both permissionless user flows (supported stablecoins ↔ USDX) and institutional collateral deposits (MBS or whole loans ↔ USDX), with tiered redemption against underlying reserves.
3. A yield-bearing, permissionless, DeFi-native staked token, mUSDX, that passes mortgage portfolio returns through to holders with exchange-rate accounting and a cooldown for unstaking.
4. A peg-stability apparatus composed of primary redemption, repo-backed liquidity, autonomous onchain market operations, and structural mortgage-repayment demand.
5. A four-step loss waterfall with strict seniority — RATES safety module slashed to exhaustion, stablecoin vault drawn to exhaustion, mUSDX exchange rate reduced up to 67% of accumulated yield, and uncapped new RATES minting as the final backstop — capitalized in the first tranche by RATES token stakers, isolating protocol-level credit and operational risk from stablecoin holders.
6. A native origination channel that allows the reserve to source newly originated mortgage collateral directly, rather than only through secondary-market purchases, aligning origination quality with reserve quality.
7. A technical architecture specification covering smart contracts, multi-provider oracle design for MBS and whole-loan pricing, and proof-of-reserves.

The remainder of this paper develops each of these in detail.

2. Background

2.1 The U.S. Mortgage Market

The U.S. residential mortgage market is one of the largest debt markets in the world, with mortgage balances of approximately $13.2 trillion outstanding as of Q4 2025 and $524 billion of new mortgage originations in that quarter alone [2]. A large majority of outstanding mortgages are either owned or guaranteed by Fannie Mae, Freddie Mac, or Ginnie Mae — the three agencies that support the conforming mortgage market. Mortgages that meet agency underwriting guidelines are typically pooled into mortgage-backed securities guaranteed by these agencies, creating agency MBS, with total agency MBS outstanding of approximately $11–12 trillion — a subset of the broader mortgage-debt total above [6]. Measured by outstanding balance, agency MBS are the second-largest sector of the U.S. bond market, behind only U.S. Treasuries; measured by trading liquidity, the market in which they trade is likewise second only to the Treasury market (the TBA market, described below). It is worth being precise about the comparison: U.S. Treasury debt outstanding (roughly $28 trillion) exceeds total mortgage debt, so the mortgage market is not larger than the Treasury market by size. What sets the underlying asset class apart is not the size of the debt but the size of what secures it — the roughly $50 trillion of U.S. residential real estate behind that $13.2 trillion of debt, an asset base far larger than the Treasury market.

The agency MBS market exists to solve a structural problem in housing finance: individual mortgages are illiquid, heterogeneous, and long-dated, which makes them difficult for originators to hold and for investors to price. By pooling conforming loans and wrapping them in an agency guarantee, the securitization model converts millions of individual mortgages into standardized, liquid, credit-risk-free instruments. This innovation, developed from the late 1960s onward, is what allows a local mortgage to be funded by global capital, and is a primary reason U.S. mortgage rates are lower and mortgage credit more widely available than in most of the world.

Agency MBS are credit-risk-free from the investor's perspective: Fannie Mae and Freddie Mac guarantee timely payment of principal and interest on securities they issue, and Ginnie Mae securities carry the full faith and credit of the U.S. government. Investors in agency MBS bear interest rate risk, prepayment risk, and liquidity risk, but not credit risk on the underlying mortgages.

Agency MBS trade primarily in two venues:

- The **TBA (To-Be-Announced) market**, where pools of MBS are traded forward based on specified characteristics (coupon, issuer, maturity) rather than specific CUSIPs. The TBA market settles monthly and is the second most liquid fixed-income market in the world after Treasuries.
- The **specified pool market**, where buyers purchase specific MBS with known underlying loan characteristics, typically at a premium to TBA pricing.

Prices update continuously during trading hours (roughly 7:00 AM to 5:00 PM Eastern) through dealer quotes and platforms such as Tradeweb and Bloomberg. After hours, pricing relies on end-of-day marks from dealers or pricing services such as ICE and Bloomberg BVAL. This price discovery mechanism is an important input to any onchain representation of MBS collateral, as discussed in Section 10.

2.2 Stablecoins Today

As of May 2026, the total stablecoin market capitalization stands at approximately $322 billion, representing more than 50% growth since early 2025 [1]. USDC and USDT together account for the majority of circulation. These instruments are dollar-pegged, primarily by holding reserves in short-duration U.S. Treasuries, overnight repurchase agreements, and bank deposits. Their yields, when passed through to holders, approximate the Secured Overnight Financing Rate (SOFR) minus operating expenses. The July 2025 GENIUS Act established the first federal regulatory framework for "payment stablecoins" — digital assets designed for payment or settlement that the issuer is obligated to redeem for a fixed monetary value and represents will hold a stable value — requiring full one-to-one backing in cash and short-duration Treasuries and prohibiting issuers from paying interest or yield directly to stablecoin holders [7].

More recent entrants have experimented with diversified yield sources. Ethena's USDe uses delta-neutral positions in perpetual futures markets to generate yield. Maker's DAI, Ondo's USDY, and several tokenized Treasury products have introduced additional reserve compositions, though all remain concentrated in short-duration government or quasi-government paper.

USDX is distinguished by its collateral composition: a portfolio of amortizing, asset-backed cashflows rather than short-duration government securities. The reserve holds longer-duration assets, but the protocol's liquidity does not depend on those assets maturing — redemption liquidity is sourced from a liquid-reserve buffer, repo facilities against agency MBS (one of the deepest short-term funding markets in the world), and a tiered redemption queue. Longer asset duration introduces interest rate and prepayment risk that must be explicitly managed, but it does not make the protocol illiquid. The point is not that mortgage collateral yields more; it is that USDX makes one of the largest asset classes in American finance — residential mortgage debt — composable. Mortgage cashflows that today reach only a narrow set of institutional allocators become the backing for a dollar any wallet can hold, move, and redeem, and that plugs natively into the rest of onchain finance.

2.3 Asset-Backed Stablecoin Prior Art

Previous attempts at asset-backed stablecoins have included commodity-backed designs (Tether Gold, PAX Gold), real-estate-tokenized designs (RealT, Lofty), and fiat-collateralized designs (USDC, USDT). None of these have combined three properties simultaneously:

1. Backed by cashflowing real-world assets at institutional scale;
2. Redeemable against reserves through an onchain mechanism;
3. Yield-bearing for holders willing to stake.

USDX targets this intersection. The closest precedent is Maker's RWA vaults, which deposit tokenized private credit as a backing component for DAI. The USDX design differs in that the mortgage collateral is the primary reserve rather than a supplementary one, and in that the yield is passed through to stakers directly rather than being captured at the protocol level.

2.4 Design Principles

The USDX protocol adheres to a set of principles that shape every component of the system:

Collateral simplicity. The portfolio is composed of a small number of well-understood asset classes with clear legal frameworks, mature price discovery mechanisms, and established custody rails. Novelty is concentrated in how these assets are represented onchain, not in what they are.

Redemption primacy. The peg is anchored by direct redeemability against reserves for $1. The protocol is designed so that a holder can always redeem USDX for one dollar of value through the redemption process. Every other peg mechanism — liquidity pools, autonomous market operations, structural repayment demand — is a supplementary arbitrage surface, not a load-bearing substitute for redemption.

Risk isolation. Credit, market, and operational risks are absorbed by mechanisms other than USDX holders wherever possible. The RATES safety module, disciplined underwriting on directly originated collateral, and staged loss absorption all exist to keep the stablecoin itself boring.

Economic separation. The economics of USDX and the protocol derive only from the reserve portfolio — the mortgage and MBS cashflows that back the stablecoin. USDX holders and the protocol do not gain from, and are not exposed to, any separate commercial activities Stable Company may conduct, such as origination services, tokenization services, or marketplace activity. The reserve backs the stablecoin; the stablecoin does not underwrite the company.

Regulatory compatibility. The system is designed to be operable within a licensed financial infrastructure stack — qualified custodians, compliant origination, audited reserves, and a licensed issuer posture. This is a deliberate posture rather than a strict technical dependency: the core mint, redeem, staking, and reserve mechanics function independently of any single jurisdiction's framework, but the protocol assumes, rather than resists, a regulated future for dollar-pegged instruments, because that is the environment in which institutional capital and mortgage collateral actually operate.

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3. System Overview

USDX is a multi-component protocol comprising four interacting subsystems:

1. The **collateral system**, which holds reserves in agency MBS, whole mortgage loans, tokenized mortgages, and liquid assets.
2. The **mint-and-redeem system**, which processes stablecoin issuance against eligible deposits and redemption against reserves.
3. The **yield system**, through which USDX holders can stake into mUSDX to receive the net yield of the underlying collateral portfolio.
4. The **token and safety system**, in which the RATES token gates new-collateral minting through validator attestations and backstops the portfolio against shortfalls through a first-loss safety module.

The operator, Stable Company, runs the off-chain components: custody relationships, mortgage servicing, and the origination channel. The onchain components — mint, redeem, staking, yield distribution, safety module, and validator attestations — are implemented as smart contracts on the deployed chains. A key design choice is the separation of off-chain operational responsibilities from onchain financial rails: the former are necessarily human and institutional, while the latter are cryptographically verifiable.

The protocol is chain-agnostic. USDX is deployed natively across multiple EVM and SVM environments, with each deployment's supply accounted for within the protocol's aggregate reserve framework.

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4. Collateral Framework

4.1 Composition

The USDX reserve portfolio is composed of four asset classes. Each operates within an allocation range rather than a fixed target; the ranges below describe steady-state operation at scale, and the portfolio may sit well outside them in early operation (notably far more liquid) as described in Section 4.1:

Asset Class	Allocation Range	Primary Function
Agency MBS	0%–80%	Credit-risk-free yield-generating reserve
Whole loans and warehouse lines	0%–60%	Higher-yield, shorter-duration component
Tokenized mortgages	0%–100%	Directly originated onchain collateral
Liquid reserves	0%–100%	Redemption liquidity and operational buffer

The ranges are wide by design. The protocol is built to operate across a spectrum of portfolio compositions reflecting both scale and market conditions. In early operation, when onchain origination is limited and MBS acquisition is ramping, the portfolio may be almost entirely liquid reserves — potentially close to 100%. As origination grows and institutional relationships mature, MBS and whole loans become meaningful components. The allocation ranges in the table describe the intended steady state, not a constraint the protocol must satisfy from day one.

Liquid reserves include supported stablecoins, short-duration U.S. Treasury bills held onchain or off-chain, balances at insured banking partners, pre-negotiated repurchase agreement capacity against MBS holdings, and positions in whitelisted onchain lending protocols that permit rapid withdrawal. What qualifies as a liquid reserve is any asset or facility that can produce redemption liquidity within one to three business days without forced sale of longer-duration collateral.

4.2 Agency Mortgage-Backed Securities

Agency MBS are a foundational reserve asset. The protocol maintains exposure to MBS issued by Fannie Mae, Freddie Mac, and Ginnie Mae, with a preference for pools with stable prepayment characteristics and coupons near current market rates.

Agency MBS are acquired through regulated broker-dealer relationships in the TBA market and the specified pool market. All securities are held by a qualified custodian in segregated accounts under the name of a bankruptcy-remote special purpose vehicle controlled by Stable Company. Stable Company does not commingle protocol assets with operating capital under any circumstances; the Articles of the SPV and associated trust documents enforce this separation.

The primary risks on agency MBS are interest rate and prepayment risk:

- Falling rates accelerate prepayments, causing MBS to pay down principal at par faster than scheduled. This creates reinvestment risk: the protocol must redeploy principal at lower yields.
- Rising rates decelerate prepayments and depress MBS market prices. The portfolio's book value declines, though held-to-maturity accounting allows the portfolio to continue earning its original coupon.

These risks are addressed by duration management, interest rate hedging where cost-effective, and transparency of the portfolio's current mark-to-market and book values to USDX and mUSDX holders via the proof-of-reserve infrastructure.

4.3 Whole Mortgage Loans

The protocol may hold an allocation to whole mortgage loans — loans owned directly by the reserve SPV rather than wrapped in an MBS structure. Whole loans typically yield 50–150 basis points above comparable MBS in ordinary market conditions, reflecting compensation for credit risk, illiquidity, and servicing complexity.

Whole loans held by the protocol are subject to underwriting and concentration guidelines that scale with the size of the portfolio:

- **Underwriting quality.** Whole loans are underwritten to agency-eligible standards or to defined non-QM standards with documented compensating factors. The protocol does not restrict itself exclusively to conforming product; it may hold agency-eligible non-QM loans where the credit profile and pricing justify inclusion.
- **Loan-to-value discipline.** For loans the protocol originates or underwrites directly, the reserve targets conservative loan-to-value ratios so that the underlying property provides an equity cushion against credit loss. This is an underwriting target for directly originated collateral, not a ceiling on every asset the reserve may hold: agency-guaranteed loans and seasoned bonds the protocol acquires may carry higher original LTVs (a conforming loan can be originated at up to 97% LTV), because on

agency collateral the credit risk is borne by the agency guarantee rather than by the protocol.

- **Geographic diversification.** As the whole-loan portfolio grows, the protocol targets geographic diversification to avoid concentration in any single state or metropolitan area. Early portfolios will necessarily be more concentrated; diversification is a target the protocol grows into as origination volume scales, not a constraint that can be satisfied at small portfolio sizes.
- **Performance requirements.** Loans must be current at acquisition. A loan that becomes 60+ days delinquent is marked down or removed from the reserve calculation and classified as non-performing until resolution.

Whole loans are acquired through forward purchase agreements with originators and through warehouse lines against which the protocol funds inventory. Warehouse lines allow the protocol to hold loans temporarily while MBS pool formation or securitization is pending.

4.4 Tokenized Mortgages

Tokenized mortgages are loans the protocol sources through a native onchain origination channel, whether originated by Stable or by licensed partners. They are represented onchain with metadata encoding origination terms, current principal balance, payment history, and lien status, and they may enter the reserve at the moment of origination.

The underlying legal instrument is a traditional note and recorded lien, governed by applicable state law. Today the onchain representation tracks that off-chain legal record; where the two differ, the recorded legal instrument governs. The mechanics of onchain mortgage origination, servicing, and the legal architecture that supports them are specified in a separate paper and are outside the scope of this document.

The relevance of tokenized mortgages to USDX is narrow and specific: they let the reserve grow with newly originated collateral sourced directly, rather than only through secondary-market purchases, which aligns the quality of newly created collateral with the quality of the reserve. For the purposes of this paper, a tokenized mortgage is simply one of the four reserve asset classes, serviced by a licensed mortgage servicer and carried in the reserve like any other whole-loan exposure.

4.5 Liquid Reserves

The protocol maintains a liquid reserve allocation sized to meet anticipated redemption flow. In steady-state operation at scale this is a minority of the portfolio, but in early operation it may be the substantial majority — the protocol does not impose a fixed ceiling on how liquid it may be. Liquid reserves are composed of any combination of:

- Supported stablecoins held in smart contracts with multi-signature governance;
- Short-duration U.S. Treasury bills, held onchain as tokenized Treasuries or off-chain via a qualified broker-dealer;
- Cash at FDIC-insured banking partners;
- Repurchase agreement capacity against MBS holdings, both committed and uncommitted;
- Positions in whitelisted onchain lending protocols that permit rapid withdrawal.

On repo capacity: a committed line is one a counterparty is contractually obligated to fund against the protocol's MBS, typically in exchange for a commitment fee; an uncommitted line is one a counterparty is expected but not obligated to fund. The protocol distinguishes the two because only committed capacity is reliable in stressed markets — precisely when it is most needed — and the protocol sizes its committed lines accordingly rather than relying on uncommitted capacity that may not be available in a crisis.

The liquid reserve serves three purposes:

1. Redemption liquidity for the default and expedited redemption paths (Section 5).
2. Operational buffer for routine protocol expenses (custody fees, servicing fees, audit costs, legal fees).
3. Duration counterweight that offsets a portion of the longer-duration exposure in the MBS allocation, smoothing the portfolio's net interest rate sensitivity.

4.6 Overcollateralization

USDX is overcollateralized, but not by a single fixed ratio. Overcollateralization is produced by the combination of three structural features:

- **Credit protection on MBS:** Agency MBS carry no credit risk to the protocol. Every dollar of par value is guaranteed by the issuing agency.
- **Underwriting discipline on directly originated collateral:** Loans the protocol originates or underwrites directly are underwritten conservatively so that the underlying property provides an equity cushion against credit loss. (As noted in Section 4.3, this discipline applies to collateral the protocol underwrites itself, not to agency-guaranteed loans whose credit risk is borne by the agency.)
- **First-loss capital:** A layered stack of first-loss capital absorbs losses before any impact flows through to USDX holders (Section 8).

The protocol targets a portfolio book-value collateralization of at least 102%. This is a target maintained through reserve management — the accumulation of mint and redemption fees, retained spread, and disciplined asset selection — not a continuously guaranteed invariant. Market events can move the ratio, which is precisely why the loss waterfall (Section 8) exists. Formally, let A_t be the book value of reserves at time t and S_t be the outstanding USDX supply at time t. The target is:

> $A_t \geq 1.02 \cdot S_t$

Sustained or severe breaches of this target trigger governance review and, where necessary, draws on the loss waterfall to restore the ratio.

4.7 Custody

All reserve assets are held in custody arrangements that meet institutional fiduciary standards:

- **Agency MBS:** qualified custodian (e.g., Bank of New York Mellon, State Street), held in segregated accounts under the name of the reserve SPV.
- **Whole loans:** held in the name of the reserve SPV, with note custody by a document custodian and servicing rights contracted to a licensed sub-servicer.

- **Supported stablecoins and onchain assets:** multi-signature smart contract wallets with a threshold signing policy, including at least one independent signer.
- **Treasury bills:** qualified broker-dealer account in the name of the reserve SPV, or tokenized Treasury positions held in multi-signature custody onchain.
- **Cash:** FDIC-insured bank accounts at Stable Company's designated banking partner(s).

The protocol publishes a monthly third-party attestation of reserve composition and value, consistent with the disclosure cadence of the largest fiat-backed stablecoin issuers, and supplements it with onchain proof-of-reserve infrastructure (Section 10) that lets anyone verify the published figures against the protocol's onchain holdings. The attestation confirms the existence, valuation, and unencumbered status of reserve assets.

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5. Mint and Redeem

5.1 Minting Surfaces

USDX supports two minting surfaces, each serving a distinct user class:

Permissionless Mint. Any user with a supported stablecoin can deposit to the mint contract and receive USDX 1:1, less a mint fee of 10 basis points. At launch the supported set begins with the most liquid, most credible dollar stablecoin and expands by governance as warranted. This surface is open, pseudonymous, and available 24/7 onchain. Deposited stablecoins are routed by the protocol treasury into the reserve portfolio, with the allocation mix determined by the current rebalancing schedule.

Institutional Mint. Institutional participants with appropriate KYC/KYB and custody arrangements may deposit eligible collateral — whole mortgage loans or agency MBS — and receive USDX at a value reflecting the mark-to-market price of the deposited assets, less a mint-side haircut. This surface is permissioned, off-chain-initiated, and settled onchain once collateral custody is confirmed by the protocol.

Both surfaces ultimately route deposits into the same reserve portfolio, and both produce the same USDX token.

5.2 Redemption

USDX redemption is structured as a tiered queue rather than an instant-settlement guarantee. The protocol offers two redemption paths at the permissionless level and a third at the institutional level. No redemption path is a same-day obligation of the protocol; industry-standard stablecoin redemption settlement is typically one to three business days, but the protocol does not commit to this schedule prior to scale.

Default redemption (7-day window, 20 bps fee). A USDX holder initiates a redemption request by depositing USDX to the redemption contract. The request is queued; redemption is processed within seven days, with the holder receiving dollars (in a supported stablecoin) at par less a 20 basis point redemption fee. This is the protocol's standard redemption path and is designed to match or improve on industry redemption practices.

Expedited redemption (1–3 business day window, 60 bps fee). For time-sensitive redemption needs, holders may elect an expedited path with a 1–3 business day target settlement window and a 60 basis point fee. Expedited redemption is best-effort: the protocol does not guarantee that sufficient rapid liquidity is available to fulfill every expedited request at every moment. Expedited requests that cannot be filled within the target window are automatically re-queued to the default 7-day path at the default fee.

Extended settlement window. In periods of elevated redemption demand or collateral market stress, redemption settlement may extend up to 45 days to allow orderly liquidation of reserve assets without forced sale at unfavorable prices. Holders are notified if their redemption request enters the extended window.

Institutional redemption. Institutional participants may redeem starting at 100,000 USDX per request against a pro rata share of the reserve portfolio, with a settlement window negotiated between the institution and the protocol and typically ranging from T+1 to T+5 business days.

The redemption-queue architecture is analogous to structures used in tokenized-credit protocols such as USDai's queue-based redemption and is consistent with how regulated asset-management products redeem against underlying portfolios. The design prioritizes peg integrity and reserve stability over the unfounded expectation of instantaneous settlement.

5.3 Fees

Mint and redemption fees accrue to the protocol and are used primarily to build reserves for USDX. Fees flow into the reserve portfolio, reinforcing collateralization over time, and may also cover operating expenses to the extent that portfolio yield alone is insufficient. Fee parameters (mint fee, default redemption fee, expedited redemption fee) are adjustable within protocol-defined bounds. Initial values are stated in Appendix B.

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6. mUSDX: Staked USDX

6.1 Motivation

Dollar-pegged stablecoins must keep their price at $1. A stablecoin that pays yield directly to its holders — by rebasing or by distributing tokens — is inferior to one that does not, because (a) it creates a moving price target, (b) it introduces tax complexity in many jurisdictions, and (c) it complicates composability with DeFi protocols that assume a fixed-unit accounting model.

The solution is a yield-bearing wrapper. mUSDX is a tokenized, non-rebasing, exchange-rate wrapper over staked USDX. Staking USDX mints mUSDX at the prevailing exchange rate; unstaking burns mUSDX and returns USDX at the updated exchange rate. The exchange rate rises monotonically as the underlying reserve portfolio generates yield. This design mirrors the successful precedent of sUSDe (Ethena), sDAI (MakerDAO), and cToken-style wrappers.

mUSDX is DeFi-native and permissionless. Any holder of USDX can stake into mUSDX and any holder of mUSDX can initiate an unstake, subject to protocol-level parameters. mUSDX composes freely with lending protocols, liquidity pools, and structured products across the ecosystem.

6.2 Mechanism

Let E_t be the mUSDX-to-USDX exchange rate at time t, with $E_0 = 1$. As the reserve portfolio accrues net yield $Y[t, t+\Delta]$ over the interval $[t, t+\Delta]$, the exchange rate updates as:

> $E_{t+\Delta} = E_t \cdot (1 + Y[t, t+\Delta] / Vstaked_t)$

where $Vstaked_t$ is the aggregate USDX value staked into mUSDX at time t.

A user staking X USDX at time t receives X/E_t mUSDX. Subsequently, the user can unstake to receive $X' = (X/E_t) \cdot E_{t+\tau}$ USDX at time $t+\tau$, where $X' \geq X$ whenever the portfolio has generated positive net yield over τ.

The exchange rate update is triggered by a keeper transaction that reads the most recent net-yield accrual from the protocol treasury contract. Update frequency is at least daily.

6.3 Unstaking Cooldown

mUSDX implements a 7-day unstaking cooldown. A user who wishes to convert mUSDX back to USDX initiates an unstake request, at which point the mUSDX is locked in the cooldown contract. After 7 days, the user may complete the unstake, burning the mUSDX and receiving the corresponding USDX.

The cooldown addresses a fundamental asset-liability mismatch. The reserve earns yield by holding assets that are not all instantly liquid — agency MBS and whole loans settle and fund on timescales of days, not seconds. A cooldown aligns the timescale on which stakers can exit with the timescale on which the protocol can convert yield-bearing assets into redemption liquidity in an orderly way, rather than through forced sales at unfavorable prices. This is the same rationale behind the cooldown on comparable staked-wrapper designs such as sUSDe.

Concretely, the cooldown serves two purposes:

1. **Orderly liquidity sourcing.** A predictable exit window lets the treasury rebalance from MBS and whole loans into liquid reserves in time to meet unstake requests, without fire-selling collateral.

2. **Run resistance.** In market stress, a cooldown dampens reflexive unstake cascades that would otherwise amplify stress on the portfolio.

During the cooldown period, mUSDX continues to accrue yield through the exchange rate mechanism. Users who need immediate liquidity may sell mUSDX in secondary markets at prevailing prices. During periods of stress, mUSDX may trade at a small discount to its exchange rate; this discount represents the market's valuation of the cooldown and is a normal feature of the design.

6.4 Yield Source

The yield accruing to mUSDX stakers consists of:

1. Agency MBS coupon income, net of custody and servicing fees.
2. Whole loan interest income, net of servicing fees and any credit losses.
3. Tokenized mortgage interest, accruing on directly originated loans held in the reserve.
4. Liquid reserve yield, reflecting the short-duration yield on supported stablecoins, Treasury bills, and whitelisted lending-protocol positions.

Minus:

- Protocol overhead allocated to reserve growth and operations.
- Any draws on the reserve associated with loss events.

The residual is distributed to mUSDX stakers through the exchange rate. USDX holders who do not stake earn no yield; their choice is between the composability and cash-equivalent behavior of USDX and the yield accrual of mUSDX.

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7. Peg Stability

The USDX peg is maintained by a layered system in which the primary mechanism — direct redemption against reserves — is supported by secondary arbitrage mechanisms that exist to minimize reliance on the primary mechanism in normal operations.

7.1 Primary Mechanism: Redemption Against Reserves

The load-bearing peg mechanism is the redemption right. Any holder of USDX can redeem to dollars through the tiered redemption structure specified in Section 5.2. If USDX trades below $1 in the secondary market, arbitrageurs profit by buying USDX at a discount and redeeming at par (net of fees). If USDX trades above $1, the mint path allows arbitrageurs to deposit a supported stablecoin and sell the minted USDX at a premium.

This is the same arbitrage mechanism that anchors every major stablecoin to its peg, and its effectiveness depends on two conditions:

1. Sufficient reserve capacity to process redemption requests on the advertised schedule.
2. Credibility of the redemption right. Users must believe, with justification, that redemptions will be honored on the stated timeline.

Both conditions are addressed by the protocol's reserve management policy (Section 4) and its public audit and proof-of-reserve infrastructure (Section 10).

7.2 Repo Facilities for Intraday Liquidity

In periods of elevated redemption demand, the protocol may exhaust its liquid reserves before redemption flow subsides. Rather than force the sale of MBS into the market — which would realize mark-to-market losses and could depress the protocol's book value — the treasury accesses repurchase agreements collateralized by its MBS holdings.

A repo transaction is a short-term borrowing against securities: the protocol delivers MBS to a counterparty in exchange for cash, agreeing to repurchase the MBS at a slightly higher price on a specified future date (typically overnight, but up to 90 days). Agency MBS repo is one of the most liquid short-term funding markets in the world, with daily volumes measured in hundreds of billions of dollars.

The protocol maintains pre-negotiated repo lines with primary dealers and bank counterparties. As described in Section 4.5, the protocol sizes its committed lines — the capacity counterparties are obligated to fund — to be sufficient for stressed conditions, rather than relying on uncommitted capacity that may evaporate in a crisis.

7.3 Liquidity Pool Arbitrage

USDX is paired with supported stablecoins in decentralized liquidity pools across its deployed chains. These pools provide a continuous two-sided market for USDX that allows price discovery and small redemption flows to occur onchain without interacting with the protocol treasury. When USDX trades off peg in these pools, arbitrageurs correct the imbalance by routing through the mint or redeem paths.

7.4 Autonomous Peg Maintenance

An autonomous peg-maintenance bot (APMB) holds deployable stablecoin capital and uses it to capture arbitrage on meaningful depegs: when USDX trades materially below par on a deployed chain's primary pool the APMB buys USDX to provide bid pressure, and when it trades materially above par the APMB mints and sells USDX to provide supply pressure. A baseline trigger of roughly ±50 basis points applies, though the bot may respond to smaller dislocations where the expected arbitrage is positive after fees.

The APMB draws its operating inventory from the Tranche 2 stablecoin staking vault (Section 8.2), so peg-defense capital scales with external staking interest rather than relying solely on operator treasury. Its activity is automatically suspended whenever the loss waterfall is active, since Tranche 2 capital is then reserved for loss absorption. The APMB is not a mandatory peg mechanism; it smooths short-term pool fluctuations, and the fundamental peg anchor remains redemption against reserves.

7.5 Dollar-Denominated Interest Demand

For the institutional mint pathway, borrowers receiving USDX against posted collateral may be required to pay interest in USDX (or in a supported stablecoin at the borrower's option). This creates a structural demand channel for USDX that is independent of speculative flows: interest obligations must be serviced regardless of the prevailing market price of USDX.

7.6 Mortgage Repayment Demand

Where the protocol holds mortgage collateral denominated in USDX, borrowers repay principal and interest in USDX at face value. Because each such loan represents a scheduled stream of future USDX-denominated payments, the aggregate originated portfolio produces a recurring, contractual demand for USDX that scales with origination volume. This is a structural demand source independent of speculative flows, and it strengthens the peg in two ways.

Counter-depeg bid. A borrower with an upcoming payment can acquire USDX on the open market; if USDX trades below $1.00, the borrower satisfies a face-value obligation with discounted tokens, capturing the discount. This incentive applies to every such borrower simultaneously and creates bid pressure on USDX whenever it trades below par, with depth that scales with the size of the originated portfolio.

Structural supply contraction. USDX received in satisfaction of principal is retired from circulation. When a borrower repays a dollar of principal in USDX, the protocol burns the received USDX and reduces the outstanding balance of the corresponding loan in the reserve by the same amount. Both the liability (USDX supply) and the asset (the loan receivable) decrease equally, preserving the collateralization ratio. Over the life of a loan, the full principal minted at origination is contractually returned to the burn contract through scheduled amortization. USDX received for interest is not burned; it flows to the mUSDX exchange rate and the protocol treasury as ordinary portfolio yield.

Importantly, denominating the loan in USDX does not make the underlying debt riskier. The borrower's obligation is a fixed dollar amount; USDX is a dollar instrument redeemable for $1, so repaying in USDX is economically identical to repaying in dollars. The borrower bears no currency or peg risk on their obligation, because the protocol stands ready to treat one USDX as one dollar of principal regardless of secondary-market price. The mechanism is analogous to the peg contribution that CDP repayment provides for crypto-collateralized stablecoins, with two differences of degree: the collateral is residential mortgage debt rather than cryptocurrency, so the repayment stream is decoupled from crypto-market volatility; and the potential scale is far larger, since U.S. mortgage originations run in the hundreds of billions of dollars per quarter.

7.7 Supported-Stablecoin Acceptance Policy

A supported mint-accepted stablecoin is automatically suspended from the mint contract if either of two conditions occurs: it trades more than 1% below par within a trailing 72-

hour window, or its issuer's redemption process is impaired such that it can no longer be reliably redeemed for a dollar. Acceptance is judged on redeemability, not price alone — a stablecoin that holds its market price but cannot be redeemed at par is as dangerous to the reserve as one trading off-peg.

Once suspended, a stablecoin is re-admitted only after it has both maintained its peg (above $0.99) and demonstrated reliable at-par redemption for a sustained 144-hour (6-day) period. The 144-hour clock begins only after the disqualifying condition has fully cleared; any recurrence resets it.

This policy prevents the protocol from absorbing devalued or unredeemable stablecoins during a crisis. During such periods, USDX remains mintable against MBS and whole-loan collateral via the institutional path.

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8. Risk Management and the Loss Waterfall

8.1 Principles

A stablecoin collateralized by a real-asset portfolio must have an explicit and credible mechanism for absorbing losses. Credit losses on whole loans, mark-to-market deficits during a forced liquidation, operational errors, or smart contract incidents can all produce a situation in which reserves fall below outstanding supply. The protocol specifies a four-step loss waterfall, operating in strict seniority, that routes losses through buffer mechanisms before they impact USDX holders.

8.2 Tranches

Losses are absorbed across four tranches, ordered from first-loss to last-resort. Each tranche operates within a coverage band — a target size expressed as a percentage of outstanding USDX supply — specified in Appendix C. The combined first-loss capacity (Tranches 1 through 3) is sized so that the protocol targets absorbing a portfolio-wide loss on the order of 30% before existing RATES holders face unbounded dilution; this target far exceeds any loss event in the modern history of agency MBS or well-underwritten whole loans.

Tranche 1 — RATES Safety Module. Staked RATES tokens in the safety module contract. The first tranche to absorb loss, and slashed to exhaustion before any subsequent tranche is touched. RATES stakers accept first-loss exposure in exchange for a share of protocol fees, RATES emissions (specified in the RATES paper), and the economic benefits of validator participation (Section 9).

Tranche 2 — Stablecoin Staking Vault. A staking vault that accepts supported stablecoins (distinct from USDX) in exchange for a yield-bearing staking receipt. Depositors earn an above-market stablecoin yield — sourced from base protocol revenue, RATES emissions, and APMB trading gains (Section 7.4) — in exchange for accepting second-loss exposure and a withdrawal delay. Tranche 2 is drawn to exhaustion after Tranche 1. When the waterfall is inactive, Tranche 2 capital is deployed as APMB working capital for peg defense; withdrawal is subject to a cooldown that compensates the protocol for the operational commitment and correspondingly elevates the tranche's yield.

Tranche 3 — mUSDX Yield Absorption. The mUSDX staking system (Section 6). mUSDX holders have explicitly accepted yield-bearing exposure to the mortgage portfolio. In a loss event, after Tranches 1 and 2 are exhausted, the protocol may reduce the mUSDX exchange rate to absorb loss — but only up to 67% of the yield accumulated in the exchange rate above 1.00. This produces a firm floor of 33% of accumulated yield below which the exchange rate cannot fall through the loss waterfall, protecting the base principal position of mUSDX holders.

Tranche 4 — New RATES Minting. Issuance of new RATES tokens that are sold to raise capital for the reserve. Tranche 4 is the final backstop and is uncapped: dilution of existing RATES holders continues until the loss is fully absorbed.

Unstaked USDX is the instrument being protected. Except in the deepest-tail scenario in which all tranches are exhausted (Section 8.4), unstaked USDX remains redeemable at par throughout any loss event. While the waterfall is active, the expedited redemption path is suspended and redemptions are processed on the default and extended schedules only, so that orderly liquidation rather than a redemption race governs the protocol's response to a shortfall.

8.3 Four-Step Loss Absorption

The waterfall proceeds in four sequential steps. Each step fires only if the prior step's absorption is insufficient to cover the remaining shortfall. There are no parallel draws, no rounds, and no buffer tranches firing alongside first-loss tranches: each tranche fires once, in order.

Step 1 — Tranche 1 slashed to exhaustion. RATES tokens in the safety module are slashed pro rata across all stakers until the safety module balance is zero or the loss is covered, whichever comes first.

Step 2 — Tranche 2 drawn to exhaustion. If Step 1 is insufficient, the stablecoin balance in the Tranche 2 vault is drawn pro rata across all stakers until the vault is empty or the loss is covered. APMB peg-defense activity is suspended for the duration of the loss event.

Step 3 — Tranche 3 exchange rate reduction. If Steps 1 and 2 are insufficient, the mUSDX-to-USDX exchange rate is reduced to absorb additional loss, up to 67% of the yield accumulated in the exchange rate above 1.00. The exchange rate cannot be reduced below a floor equal to 33% of its accumulated yield above 1.00, protecting the base principal position of mUSDX holders.

Step 4 — Tranche 4 uncapped RATES minting. If Steps 1, 2, and 3 together are insufficient, all further absorption is funded by uncapped minting of new RATES tokens, which are sold into the market with proceeds applied to the reserve. Dilution of existing RATES holders is the final backstop and continues until the loss is fully absorbed or wind-down mode activates (Section 8.4).

Figure 1 illustrates the sequence; Table 1 summarizes the absorption caps.

Figure 1: USDX four-step loss waterfall. Each step fires only if prior steps are insufficient to cover the remaining shortfall.

Step	Tranche	Action	Cap
1	T1 — RATES safety module	Slash staked RATES pro rata	To exhaustion
2	T2 — Stablecoin vault	Draw vault balance pro rata	To exhaustion
3	T3 — mUSDX yield	Reduce mUSDX exchange rate	Up to 67% of accumulated yield
4	T4 — New RATES mint	Mint and sell new RATES	Uncapped

Table 1: USDX four-step loss waterfall. Each step fires only if prior steps are insufficient.

Under this structure, the aggregate loss absorption available before unbounded RATES dilution activates is the full T1 staked supply, the full T2 vault balance, and up to 67% of mUSDX accumulated yield. This buffer is sized to absorb losses at a scale substantially beyond historical precedent in the agency MBS or well-underwritten whole loan universe.

8.4 Deep-Tail Scenario

Tranches 1 through 3, together with Step 4's uncapped mint, are designed to cover a portfolio-wide loss well beyond the protocol's ~30% first-loss target — a magnitude with no precedent in the modern history of agency MBS or well-underwritten whole loans. If losses exceed even Step 4's practical market capacity — that is, if the market cannot absorb the necessary volume of newly minted RATES at any meaningful price — the protocol enters wind-down mode: new mints are halted and remaining reserves are distributed pro rata to redeeming USDX holders. This is the only scenario in which USDX holders receive less than $1 per token.

8.5 Recovery

When reserves return to parity and protocol operations normalize, a recovery waterfall makes slashed tranches whole in the inverse order of their loss sequence — last-in, first-out. Tranche 3 yield is restored first, then Tranche 2 vault balances are recapitalized, then Tranche 1 safety module stake is rebuilt. This preserves the incentive for first-loss participants to remain in position across cycles.

8.6 Specific Risk Categories

Interest rate risk. Addressed by duration management of the MBS portfolio, selective hedging through interest rate swaps or Treasury futures where cost-effective, and by maintaining liquid reserves that offset long-duration exposure.

Prepayment risk. Addressed by holding a diversified pool of MBS with varying coupon and seasoning characteristics, and by the structural ability to redeploy prepaid principal into new MBS or whole loans at prevailing market rates.

Credit risk. Limited by the agency guarantee on MBS (which bears the credit risk on agency collateral), by conservative loan-to-value discipline on collateral the protocol originates or underwrites directly, and by the four-step loss waterfall. The protocol does not assume the underwriting standards of collateral it does not itself originate; agency-guaranteed exposure carries no protocol credit risk regardless of the underlying loan's original LTV.

Smart contract risk. Addressed by security audits from multiple independent firms and staged deployment with time-locked parameter changes.

Oracle risk. Addressed by redundant oracle sources across multiple providers, price deviation guards, and circuit breakers that halt protocol operations during oracle anomalies. See Section 10.

Operational risk. Addressed by qualified custody, multi-signature authorization for treasury operations, and continuity planning.

Regulatory risk. Addressed by proactive engagement with regulators, compliance-first product design, KYC/AML at all origination and institutional-facing surfaces, and a licensed posture for all regulated activities (see Section 12).

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9. The RATES Token

RATES is a component of the USDX protocol rather than a standalone product. This paper describes RATES only to the extent necessary to understand USDX. A separate RATES paper specifies distribution, supply schedule, and detailed tokenomics.

RATES has three core utilities within the USDX system:

9.1 Validator Staking

Validators stake RATES to join the active validator set. Validators are responsible for two classes of attestations that gate the minting of new USDX collateral:

- **Origination attestations.** At the origination of a new tokenized mortgage, whether sourced through Stable's own channel or an institutional partner pipeline, validators attest that loan data, appraisals, and title documentation supplied by approved third-party providers meet protocol underwriting requirements. Threshold signatures from the active validator set are required for the new loan to be admitted to the USDX reserve portfolio.
- **Quarterly attestations.** On a quarterly cadence, validators re-attest property valuation and lien status for tokenized mortgages in the portfolio, ensuring that the onchain representation remains synchronized with the off-chain legal reality.

Scope of attestation. Validators attest to the cryptographic integrity of loan data produced by approved third-party providers — AVM vendors, title companies, credit bureaus, flood and tax data providers — rather than independently producing that data. The validator's role is verification-of-verification: confirming that the data feeds supplying a new or existing loan are correctly signed, provenanced, and compliant with protocol underwriting requirements. Independent production of underlying data (appraisals, title searches, credit pulls) remains the responsibility of the licensed service providers that perform those functions today.

Economic structure. Validators earn modest flat or bps-capped fees per origination and per quarterly re-attestation, priced to be materially below the combined cost of the traditional services they replace (quality control, document custody, and ongoing portfolio oversight). All validator fees are paid in USDX. A portion of each validator fee (specified in the RATES paper) is routed to the RATES buyback-and-burn contract; the balance flows to the validator. Validator fees are structured as cost-recovery for the attestation work performed; the primary compensation for operating a validator is the RATES emissions distribution specified in the RATES paper, supplemented by safety module rewards earned on staked RATES (Section 9.2).

Set composition. Validating activates alongside the native origination channel (Roadmap Phase 2), not at stablecoin launch, since attestations are only required once the protocol originates collateral directly. The initial active validator set is sized for threshold-signature robustness — targeted at 21 validators at origination launch and scaling toward a steady-state cap as the protocol matures, with a minimum self-stake per seat and support for delegation. Specific selection, self-stake, delegation, and slashing parameters are specified in the separate RATES paper and finalized ahead of origination launch.

Validating and first-loss are distinct. Staking RATES to validate is a separate activity from staking RATES into the first-loss safety module (Section 9.2). A validator earns attestation fees and emissions for verification work; a safety-module staker accepts loss exposure in exchange for first-loss rewards. A participant may do one without the other. Operating a validator does not, by itself, place a validator's stake at first-loss risk in the loss waterfall.

9.2 First-Loss Safety Module

Separately from validating, RATES holders may stake into the safety module — the senior-most tranche of the USDX loss waterfall (Section 8). If the reserve experiences losses it cannot absorb itself, safety-module stake is slashed pro rata to make USDX holders whole before any other tranche is affected. This is a deliberate acceptance of first-loss exposure, and it is what distinguishes a safety-module staker from a validator.

Compensation for first-loss exposure shifts as the protocol matures. In the bootstrap phase, before fee revenue scales, safety-module rewards are paid primarily in RATES emissions (specified in the RATES paper). As origination and protocol revenue grow, a share of protocol fees supplements and eventually predominates over emissions. The intent is that first-loss capital is always compensated, with the source of that compensation migrating from token emissions toward real fee revenue over time.

9.3 Governance

RATES is intended to carry governance rights over a deliberately narrow set of protocol parameters — matters such as approved data vendors, approved counterparties, and risk parameters — rather than broad control of the protocol. The specific governable parameters, vote-weighting method, and procedures are defined in the separate RATES paper and may evolve ahead of token distribution.

Governance scope is intentionally limited. USDX is designed to operate within a licensed financial infrastructure stack, and operational decisions — custody relationships, servicing contracts, regulatory interactions — are retained by Stable Company as the licensed operator. RATES governance is complementary to, not a replacement for, the regulated operating entity.

9.4 Emergency Response

As the licensed operator, Stable Company retains a narrow set of emergency powers — pausing mints, pausing redemptions, halting the APMB, or triggering a loss-waterfall draw — to respond to active incidents. Stable Company discloses material emergency actions as appropriate and as required by its regulatory and contractual obligations, exercising operational discretion over the timing and detail of disclosure during a live incident.

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10. Technical Architecture

10.1 Smart Contract System

The protocol's smart contract system consists of the following core modules, each deployed on every supported chain:

Mint/Redeem Contract. Processes USDX issuance and redemption against supported-stablecoin deposits on the permissionless path. Maintains the tiered redemption queue (default, expedited, extended).

Treasury Contract. Manages onchain reserve holdings, triggers rebalancing operations, and routes yield income to the Distributor Contract.

Staking Contract. Handles USDX → mUSDX deposits and mUSDX → USDX withdrawal requests. Maintains the USDX/mUSDX exchange rate and manages the cooldown queue.

Distributor Contract. Distributes protocol fees to RATES validators and the RATES safety module, updates the mUSDX exchange rate with the allocated share, and routes the balance to the Treasury.

RATES Safety Module. Staking contract for RATES tokens. Governs slashing events in response to loss waterfall triggers and recovery flows.

Validator Contract. Maintains the active validator set, processes origination and quarterly attestations, and aggregates threshold signatures gating new-collateral minting.

APMB Contract. Autonomous peg-maintenance bot with locked parameters and a pre-funded inventory.

Proof-of-Reserve Adapter. Reads the off-chain reserve attestation and publishes an onchain hash that can be verified against the published attestation document.

All contracts are upgradeable through a time-locked governance mechanism. Upgrade paths are restricted to parameter changes and security patches; logic upgrades that materially alter the protocol's economic behavior require extended timelock.

10.2 Oracle Design

The protocol relies on oracle feeds for three purposes: stablecoin price monitoring (for the stablecoin acceptance policy), agency MBS pricing (for portfolio valuation), and whole loan valuation (for book-value accounting). Oracle feeds are sourced from multiple leading providers, including Chainlink, Pyth, RedStone, Switchboard, and other established price-feed networks. The protocol does not depend on any single provider.

- **Stablecoin oracles.** Sourced across multiple providers with a median aggregation and a deviation guard that triggers circuit-breaker behavior if any source disagrees with the median by more than 1%.
- **Agency MBS pricing.** Sourced from a consortium of institutional pricing services (ICE, Bloomberg BVAL) and published onchain via a Stable-operated oracle adapter with multi-signer attestation. MBS prices update intraday during trading hours and end-of-day via dealer mark. The oracle adapter includes sanity bounds: no single update may move a CUSIP's price by more than 2% relative to the prior update without a second independent attestation.

- **Whole loan valuation.** Whole loans are carried at book value for protocol accounting, marked down only in the event of delinquency or credit events. A quarterly independent valuation serves as a check on book-value accounting and is published alongside the reserve attestation.

The oracle architecture is deliberately conservative. MBS and whole loan valuations are primary inputs to the invariant that governs mint capacity (Section 5.4); any oracle manipulation or data error could lead to over-minting. The multi-source, attestation-gated, bounded-update architecture ensures that oracle anomalies surface as pauses rather than as silent mispricings.

10.3 Proof of Reserve

The proof-of-reserve system produces a daily onchain attestation of the reserve portfolio. The attestation includes:

- Aggregate book value of reserves.
- Decomposition by asset class (MBS, whole loans, tokenized mortgages, liquid reserves).
- Outstanding USDX supply across all deployed chains.
- The collateralization ratio A_t/S_t.
- A cryptographic commitment to the detailed reserve schedule, verifiable against an off-chain published schedule.

The attestation is signed by Stable Company's treasurer and an independent auditor, both of whom maintain hardware-secured signing keys. The onchain hash of the attestation serves as the canonical reserve-status oracle for all protocol operations.

Consistent with Section 4.7, an independent third party issues a monthly attestation of the reserve portfolio, covering existence, ownership, valuation methodology, and encumbrance status. The current reserve composition, collateralization ratio, and published attestations are available on the protocol's live reserves page at trystable.co/reserves, where anyone can verify the published figures against the protocol's onchain holdings.

11. Native Origination Channel

In addition to acquiring collateral in secondary markets, USDX can source newly originated mortgage collateral directly through a native onchain origination channel. Loans originated through this channel may be funded and denominated in USDX, and may enter the reserve at the moment of origination, which aligns the quality of newly created collateral with the quality of the reserve and lets the reserve grow with origination rather than only through secondary purchases.

Origination follows the same controls as any reserve collateral: underwriting inputs include automated valuation model outputs from multiple vendors, title search, credit report, and verification of insurance and tax status; origination attestations from the RATES validator set (Section 9.1) are required before originated collateral enters the reserve; and the underwriting decision and credit file are retained by the licensed mortgage originator and available to regulators upon request. The legal instrument is a traditional note and recorded lien governed by applicable state law.

The consumer-facing product through which Stable sources directly originated collateral — including its account structure, the homeowner-facing liquidity features it enables, and its servicing and payment architecture — is specified in a separate product paper and is outside the scope of this document. For the purposes of USDX, what matters is narrow: directly originated loans are a source of reserve collateral, underwritten and attested to the same standard as any other asset the reserve holds. Where a loan is funded and denominated in USDX, its scheduled repayments contribute to the structural demand described in Section 7.6; others may be funded in USDC or another supported stablecoin.

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12. Regulatory Framework

12.1 Regulatory Positioning

USDX is designed as a payment stablecoin: a redeemable, fully collateralized digital dollar issued by a regulated entity, or chain of regulated entities, holding the appropriate state and federal registrations. Its reserves are held under the custody and bankruptcy-remote arrangements described in Section 4.7; its redemption right is a stated and contractually enforceable obligation; and its reserves are subject to the periodic third-party attestation described in Section 10.3. The protocol's compliance posture — licensed issuance, qualifying reserves, an enforceable redemption right, periodic audit and attestation, and consumer-protection disclosures — is maintained consistent with the applicable federal and state regulatory frameworks for dollar-denominated digital instruments as they are finalized.

12.2 KYC/AML and Sanctions Compliance

The protocol's institutional mint and redemption paths are subject to full KYC/KYB, AML monitoring, and sanctions screening. The permissionless mint path (supported stablecoin → USDX) is not KYC-gated at the transaction level but is subject to sanctions screening at the address level (blocking known OFAC-listed addresses). Mint, redemption, and staking are disabled for addresses flagged by the screening service.

Mortgage origination conducted through the protocol's native channel is subject to full KYC, income verification where applicable, and compliance with the Truth in Lending Act, Real Estate Settlement Procedures Act, and all applicable state lending laws.

12.3 Securities Considerations

USDX is designed as a payment stablecoin and is not intended as a security. It does not provide any claim on profits, governance rights, or residual value. Its value is limited to the redemption right, and its marketing and disclosure language reflect this positioning.

mUSDX is a DeFi-native, permissionless yield token. It can be minted and redeemed by any holder of USDX and circulates as a standard yield-bearing asset within DeFi composability layers — lending protocols, liquidity pools, and structured products — in the same mold as sDAI, sUSDe, and similar staked wrappers. mUSDX distribution or marketing may be limited in certain jurisdictions based on local regulatory treatment of yield-bearing instruments, but the token itself is a native permissionless DeFi primitive, not a gated product.

RATES is a staking and governance token whose characterization is jurisdiction-specific. Distribution is subject to the applicable regulatory framework in each deployment jurisdiction.

12.4 Custodial and Banking Posture

The protocol's banking and custody relationships are structured to meet institutional fiduciary standards. Stable Company maintains qualified custodian relationships for MBS and Treasury holdings, bank partner relationships with FDIC-insured institutions for cash balances, and licensed mortgage originator and servicer relationships for its origination and servicing operations. A full compliance stack — corporate governance, risk management, IT security, business continuity — is maintained to institutional standards.

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13. Roadmap

The protocol is developed in phases, each phase concluding with a protocol-level milestone that is publicly attested and audited.

Phase 1 — Foundation. Mainnet deployment of the core mint/redeem/staking contracts; initial reserve portfolio composed primarily of supported stablecoins and short-duration Treasuries; launch of USDX liquidity pools; APMB and proof-of-reserve infrastructure live.

Phase 2 — Origination Launch. Activation of the native origination channel in initial U.S. states (subject to licensing and underwriting readiness); first directly originated tokenized mortgages entering the reserve portfolio at the moment of origination, with origination attested operator-side during this phase.

Phase 3 — Validator Set Activation. Stand-up of the decentralized RATES validator set, transitioning origination and quarterly attestations from operator-side verification to threshold-signed attestation by staked validators (Section 9.1).

Phase 4 — Institutional Collateral. Scaling of agency MBS holdings into the reserve; onboarding of first whole loan portfolios; activation of institutional mint and redemption paths; first audited attestation covering all four reserve asset classes.

Phase 5 — Safety Module Activation. Distribution of RATES token; activation of the RATES first-loss safety module; introduction of the Tranche 2 stablecoin staking vault.

Phase 6 — Cross-Chain Expansion. Deployment of USDX to additional chains, with each deployment's supply separately backed within the aggregate reserve accounting framework.

Phase 7 — Institutional Distribution. Integration with institutional distribution partners (asset managers, fintech distributors, broker-dealers) for USDX and mUSDX; exploration of regulated product wrappers.

Concrete dates are not specified here because regulatory, partnership, and technical dependencies make them dependent on factors outside Stable Company's sole control. The protocol's public documentation maintains a current status view of each phase.

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14. Related Work

Fiat-backed stablecoins. USDC, USDT, and similar instruments have established the pattern of a regulated issuer backing a dollar-pegged token with short-duration government securities. USDX shares the redemption primitive and the regulatory posture, but departs by holding longer-duration, cashflowing collateral.

Crypto-collateralized stablecoins. MakerDAO's DAI, Liquity's LUSD, and related designs back their stablecoins with crypto collateral using overcollateralized CDPs. These designs achieve decentralization at the cost of capital efficiency, and have historically struggled in market stress events where collateral and liability are correlated. USDX's collateral is structurally uncorrelated with crypto markets, addressing this failure mode.

Algorithmic stablecoins. Terra's UST, Iron Finance's IRON, and prior algorithmic designs attempted to maintain a peg without full collateralization. The failure modes of these designs are well-documented; USDX does not pursue this design and is fully collateralized in all operational states.

Delta-neutral yield stablecoins. Ethena's USDe backs its stablecoin with delta-neutral positions in crypto perpetual futures, generating yield from funding rate spreads. This shares the staked-wrapper pattern (sUSDe) with USDX's mUSDX, but differs fundamentally in collateral — USDe is backed by derivatives positions while USDX is backed by real-asset cashflows.

Tokenized real-world assets. Ondo's USDY, Backed Finance's products, and Maker/Sky's RWA vaults have introduced tokenized T-bills and private credit to the onchain ecosystem. USDX extends this trajectory into the residential mortgage asset class, the largest single RWA category by outstanding principal.

Hardware-backed stablecoins. USDai backs its synthetic dollar with tokenized AI hardware collateral and introduces a queue-based redemption mechanism. USDX shares the queue-based redemption architecture while applying it to a different collateral class — residential mortgages rather than GPU infrastructure.

Onchain mortgage credit. Recent efforts have begun directing onchain capital toward residential mortgage credit — most visibly through private-credit facilities and warehouse-style funding lines that channel stablecoin-ecosystem capital to mortgage originators, such as arrangements connecting the Sky ecosystem with originators like Better. These structures principally provide a funding channel: onchain capital funds mortgage lending, with the loans warehoused or held on the originator's side. USDX differs in architecture rather than degree. Rather than supplying a credit line to a third-party lender, USDX holds mortgage and MBS collateral directly in its own reserve, can source newly originated loans into that reserve through a native origination channel, diversifies across four reserve asset classes including agency MBS, passes portfolio yield to holders through mUSDX, and protects holders through an explicit four-step loss waterfall. The distinction is between funding someone else's mortgage book and building the stablecoin's own reserve out of mortgage collateral.

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15. Conclusion

The U.S. dollar, as an institutional instrument, is supported by a broad and productive collateral base in which residential real estate debt plays a leading role. Stablecoins, in their current incarnation, replicate only a narrow slice of that collateral base. USDX closes the gap by backing a dollar-pegged token with the same asset class — residential mortgages and agency MBS — that has underwritten the American dollar for more than half a century.

This is a deliberate choice of foundation. A stablecoin meant to serve as a unit other systems can build on, settle in, and hold without a second thought must rest on collateral that is durable, deep, and proven across cycles. Short-duration government paper is sound but narrow; the residential mortgage and agency MBS markets are among the largest, most liquid, and most rigorously regulated asset pools in the world, and they have funded American housing through every market regime of the past fifty years. Building USDX on that base is what lets it behave less like a product and more like infrastructure: something durable enough that others can depend on it.

The design choices specified in this paper — a disciplined collateral framework, dual mint-and-redeem pathways with a tiered redemption queue, a permissionless yield-bearing staked wrapper, a layered peg stability apparatus, and a four-step loss waterfall operating in strict seniority with first-loss RATES backing — all serve that single end: a stablecoin that is redeemable, auditable, and productive by construction, usable by institutions and individuals alike without forcing either to compromise their operational or risk standards.

USDX is not a DeFi-native experiment. It is bedrock — designed to operate within the regulated financial stack while exporting the cashflows, liquidity, and composability of that stack to anyone, anywhere.

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16. References

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17. Appendix A — Glossary

Agency MBS. Mortgage-backed securities issued or guaranteed by Fannie Mae, Freddie Mac, or Ginnie Mae.

APMB. Autonomous Peg Maintenance Bot. An onchain process operated by Stable Company that trades USDX against liquidity pools to smooth meaningful peg deviations.

Loss Waterfall. The four-step ordered sequence (T1 → T2 → T3 → T4) in which losses are absorbed by different protocol tranches in strict seniority before impacting USDX holders.

mUSDX. The non-rebasing, exchange-rate staked version of USDX, accruing the net yield of the reserve portfolio. DeFi-native and permissionless.

Open-End Deed of Trust. A recorded real estate lien securing a revolving obligation up to a specified maximum, used as the legal anchor for directly originated mortgage collateral.

RATES. The protocol's validator, safety-module, and governance token.

Safety Module. The staking contract in which a portion of RATES tokens is locked to backstop the protocol against collateral shortfalls (Tranche 1 of the loss waterfall).

TBA Market. To-Be-Announced market. The primary forward market for agency MBS.

USDX. The protocol's USD-pegged stablecoin.

Whole Loan. A mortgage loan held directly, rather than as part of an MBS pool.

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18. Appendix B — Key Formulas and Parameters

Collateralization Target.

> $A_t \geq 1.02 \cdot S_t$

Where A_t is the aggregate book value of reserves at time t, and S_t is the outstanding USDX supply.

mUSDX Exchange Rate Update.

> $E_{t+\Delta} = E_t \cdot (1 + Y[t, t+\Delta] / Vstaked_t)$

Initial Parameter Set.

Parameter	Initial Value
Permissionless mint fee	10 bps
Default redemption window	7 days
Default redemption fee	20 bps
Expedited redemption window	1–3 business days (best effort)
Expedited redemption fee	60 bps
Extended redemption window (max)	45 days
Institutional redemption minimum	100,000 USDX
mUSDX unstaking cooldown	7 days
APMB depeg trigger (baseline)	± 50 bps ($0.995 / $1.005)
Single-transaction cap	2% of supply
Agency MBS allocation range	0% – 80%
Whole loans / warehouse lines range	0% – 60%
Tokenized mortgages range	0% – 100%
Liquid reserves range	0% – 15%
Whole loan and tokenized mortgage LTV cap	85%
Stablecoin depeg threshold	1% below par
Stablecoin suspension window	72 hours
Stablecoin recovery window	144 hours
Loss waterfall Step 1 (T1) cap	To exhaustion
Loss waterfall Step 2 (T2) cap	To exhaustion
Loss waterfall Step 3 (T3) cap	67% of accumulated mUSDX yield
mUSDX exchange-rate floor in waterfall	33% of accumulated yield above 1.00

Parameter	Initial Value
Loss waterfall Step 4 (T4) cap	Uncapped

All parameters are subject to adjustment within protocol-defined ranges through the governance mechanism specified in Section 9.3.

19. Appendix C — RATES Reference

RATES is the protocol's validator, safety-module, and governance token. Its full tokenomics — initial supply, distribution, emissions schedule, activity gating, buyback-and-burn, permanent protocol capital, and supply dynamics — are specified in the separate **RATES paper**. This appendix carries only the parameters the USDX body references directly.

Genesis supply. RATES has a fixed initial supply of **1,000,000,000 tokens (1B)** at genesis. Additional supply is created post-genesis only via the loss waterfall (Section 8) and a geometrically decaying emissions schedule; the schedule and its asymptotic supply cap are specified in the RATES paper.

Tranche coverage bands. Each loss-waterfall tranche (Section 8.2) operates within a coverage band, expressed as a percentage of outstanding USDX supply:

Tranche	Minimum coverage	Maximum coverage
Tranche 1 — RATES safety module	0.5%	10%
Tranche 2 — Stablecoin vault	1%	10%
Tranche 3 — mUSDX	2%	10%

The mechanics governing how tranche emissions split between staker rewards and protocol-owned capacity building, the emissions split across validators and tranches, and all other supply dynamics are specified in the RATES paper.

This whitepaper is a technical description of the USDX protocol as designed. It does not constitute an offer to sell, or the solicitation of an offer to buy, any security or financial instrument. Any forward-looking statements reflect the current intentions of the protocol's developers as of the publication date and are subject to change based on technical, regulatory, and market developments. Prospective users should consult the latest protocol documentation and, where applicable, their own legal and financial advisors.

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